



BEST PRACTICE SERIES

Using Idle Time VIMS™
Data for Pre-PM Analysis

Maintenance and Repair Application Component Renewal (CRC) Component Life Management MARC Management

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1.0 Introduction

Idle time VIMS data can be used during pre-PM analysis to speed the pre-PM process and check systems that may otherwise have gone unchecked. The idle time data can be taken while no one is present at the machine, or while inspection personnel are doing other checks. Examining 5 minutes of idle time data can eliminate the need for more time consuming tests, such as checking the steering cycles with gauges, and can detect faults in systems that may have gone unchecked, like the battery and charging system.

2.0 Best Practice Description

5 minutes of idle and 1 minute of engine off data collected during the Pre PM inspection can be used to speed up the Pre PM analysis. The collected data can be analyzed for certain truck performance parameters without operator influences. This analysis can allow the PM downtime to be utilized more effectively.

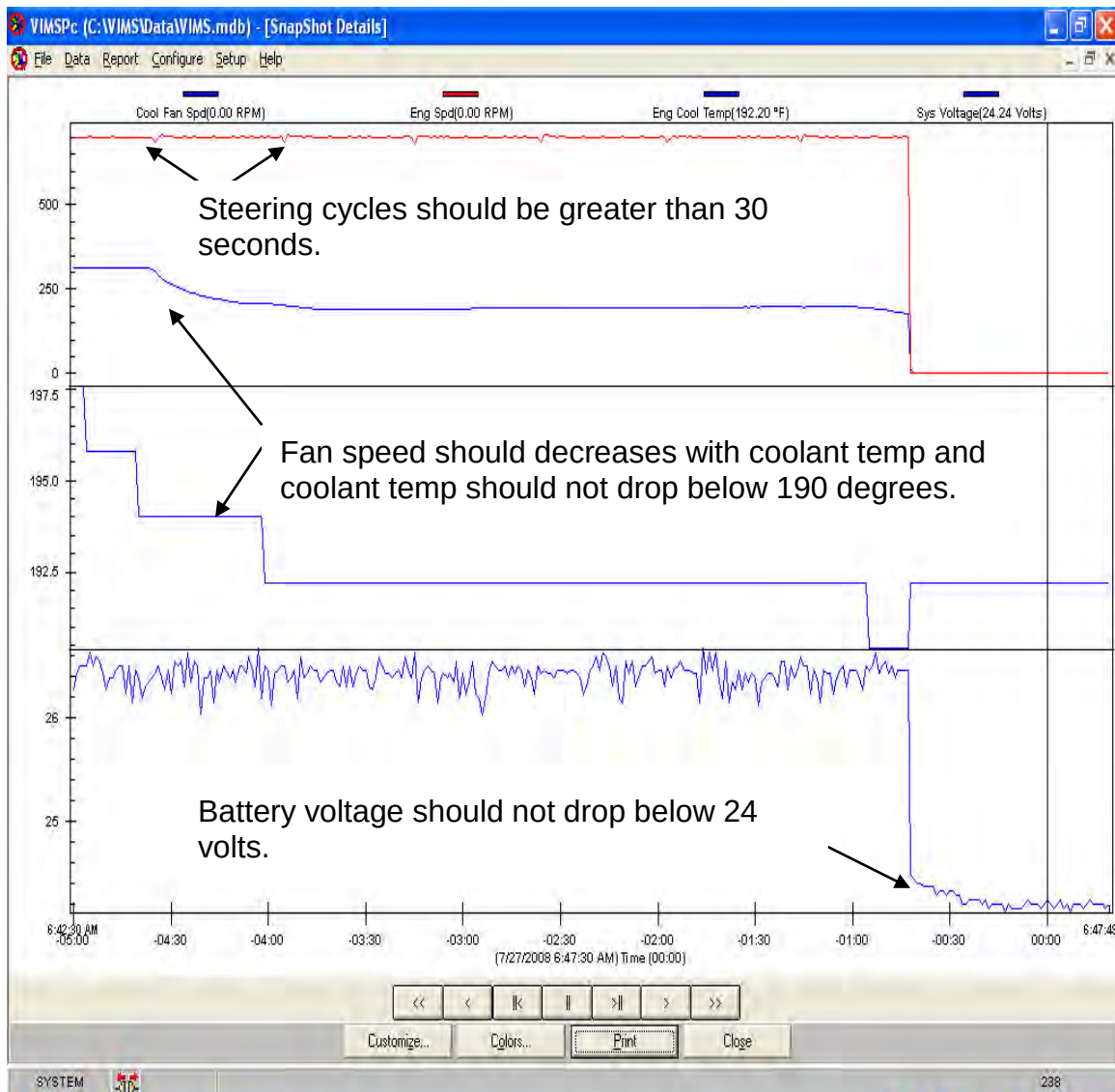
Assuming the truck is shut off, start the truck, run through operational tests and inspect machine while it idles for at least 5 minutes before shutting the machine down. After the engine is stopped, turn to key on, take a snapshot (EREC) and download the data before releasing the truck. The same data can be gathered by running a datalogger for the entire pre-PM test/inspection. Note: If an aftermarket shutdown timer is available, it can be used to collect the same data.

The downloaded VIMS® data can then be analyzed for the following:

- **Battery voltage** – Battery voltage is an excellent indicator of the condition of the charging system when the truck was running and the battery condition when the engine was shut down. When the engine is off and key on a voltage drop below 24 volts is indicative of a problem in the battery system.
- **Engine fan speed & coolant temperature** – This data can help insure the engine fan control and cooling circuit are working properly and not over cooling the engine. If the steering cycle blips can be seen in the engine fan speed graph, or the engine coolant temperature drops below 190 degrees the fan has likely locked up and is over cooling the engine.
- **Engine speed** – A blip in the engine speed can be seen every time the steering cycles. This data eliminates the need to check steering cycles with gauges and the need to check accumulators if the cycle times are correct. Cycle times less than 30 seconds can indicate a cut in or a cut out problem in the steering system.
- **Electrical problems** – The 5 minutes of continuous information from all sensors on the truck can be useful in troubleshooting should there be an electrical problem or intermediate event on the truck. Also, if a sensor is going bad it can be added to the PM kit.

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3.0 Implementation Steps

- Three days prior to the scheduled PM let the machine idle for 5 minutes while doing other pre-PM inspections, download the data before releasing the machine.
- Analyze the data and modify the PM appropriately based on findings.

4.0 Benefits

- Saves time by allowing systems to be checked through VIMS™ data rather than by hooking up gauges.
- Improves PM efficiency by providing information that may detect problems that would have been overlooked during normal pre-PM inspections.
- Can help improve fleet availability.

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5.0 Resources Required

The major resource required to implement this Best Practice will be the people and the time dedicated to analyzing the data collected from the machine. The VIMS data will be collected from the machine using the VIMSpC software. VIMS Application training may also be required if the users are not familiar with the analysis of VIMS data.

6.0 Supporting Attachments / References

References:

VIMSpC subscription numbers: JERD2137 (Dealer Site License)
JERD2138 (Dealer Annual Subscription)
JERD2175 (Customer Annual Subscription)

VIMS Application Guide SELD7001

7.0 Related Best Practices

[1006-2.3-1027](#) [Utilizing VIMS Data for Effective Planned Maintenance](#)

8.0 Acknowledgements

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